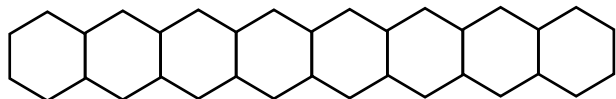


Series : ZYWX1

SET ~ 1

रोल नं.

Roll No.



कोड नं.
Q.P. Code

56/2/1

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.



CHEMISTRY

Haloalkanes and Haloarenes



निर्धारित समय: 2 घंटे

Time Allowed : 2 hours

अधिकतम अंक : 50

Maximum Marks : 50

नोट / NOTE

- (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 7 हैं।
Please check that this question paper contains 7 printed pages.
- (II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 25 प्रश्न हैं।
Please check that this question paper contains 25 questions.
- (III) प्रश्न-पत्र में दाहिने हाथ की एयर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख्य-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book at the given place before attempting it.
- (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.



General Instructions:

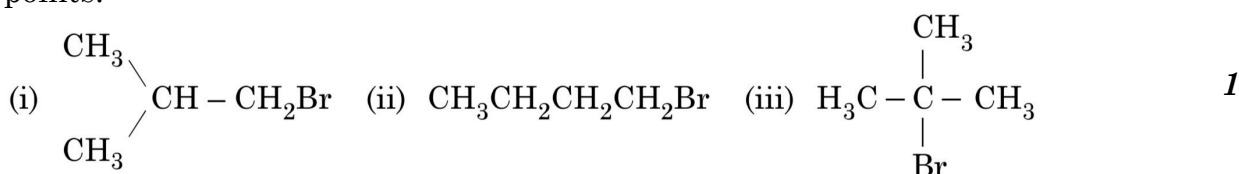
Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **25** questions. **All** questions are **compulsory**.
- (ii) This Question paper is divided into **FIVE** sections- Sections **A, B, C, D** and **E**.
- (iii) In **Section A** : Question numbers **1** to **13** are Multiple Choice (MCQ) type questions. Each question carries **1** mark.
- (iv) In **Section B**: Question numbers **14** to **17** are Very Short Answer (VSA) type questions. Each question carries **2** marks.
- (v) In **Section C**: Question numbers **18** to **22** are Short Answer (SA) type questions. Each question carries **3** marks.
- (vi) In **Section D** : Question number **23** is a Case Study-Based question carrying **4** marks.
- (vii) In **Section E** : Question numbers **24** and **25** are Long Answer (LA) type questions. Each question carries **5** marks.
- (viii) There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.
- (ix) Kindly note that there is a separate question paper for Visually Impaired candidates.
- (x) Use of calculators is **NOT** allowed.

SECTION - A

This section contains **13** Multiple Choice Questions carrying **1** mark each.

1. Arrange the following compounds in the increasing order of their boiling points:



The correct order is:

- (A) (ii) < (i) < (iii) (B) (i) < (ii) < (iii)
(C) (iii) < (i) < (ii) (D) (iii) < (ii) < (i)
2. Which of the following haloalkanes reacts with aqueous KOH most rapidly by $\text{S}_\text{N}1$ reaction? 1
- (A) 2, 2- Dimethyl-1-Chloropropane (B) 2-Chlorobutane
(C) 2-Bromo-2-Methylpropane (D) 1-Bromobutane



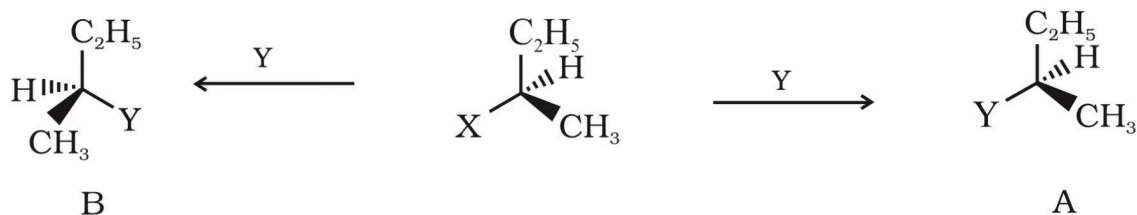
3. The conversion of an alkyl halide into an alkene by alcoholic KOH is classified as 1
- (A) An addition reaction (B) A dehydration reaction
(C) A substitution reaction (D) A dehydrohalogenated reaction
4. Chloramphenicol is used in the treatment of which of the following? 1
- (A) Typhoid fever (B) Goiter
(C) Malaria (D) Dengue
5. Plane polarised light is passed through A and is rotated when it is passed through a solution of compounds B and C. The angle of rotation is measured by an instrument D. The solution B is found to be dextrorotatory and rotates the plane of plane polarised light in a direction X, while the solution C is leavo-rotatory and rotates the plane towards direction Y. A, D, X and Y, respectively are: 1
- (A) Optical prism, Refractometer, anti-clockwise, clockwise
(B) Nicol prism, Polarimeter, clockwise, anti-clockwise
(C) Nicol prism, Polarimeter, anti-clockwise, clockwise
(D) Optical prism, Spectrophotometer, clockwise, anti-clockwise
6. Which of the following molecules is chiral in nature? 1
- (A) 2-chlorobutane (B) 1-chloropropane
(C) 2-chloropropane (D) 1-chlorobutane
7. Racemisation occurs in 1
- (A) S_N2 reaction (B) Neither S_N2 nor S_N1 reactions
(C) S_N2 and S_N1 reactions (D) S_N1 reactions
8. Which of the following compounds would be hydrolysed by aqueous KOH most easily? 1
- (A) $CH_2 = CH - Br$ (B) $CH_2 = CH - CH_2 - Br$
(C) $CH_3 - CH(Br) - CH_3$ (D) $CH_3 - CH_2 - Br$
9. Auto-oxidation of chloroform in air and light produces a poisonous gas know as
- (A) Mustard gas (B) Phosphine



(C) Phosgene

(D) Tear gas

10. Consider two compounds A and B formed from a common reactant:



1

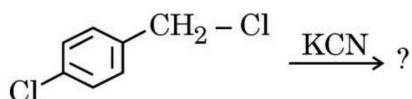
If a 50:50 mixture of A and B is obtained, then consider the following statements:

- (i) Racemisation process occurs and product is optically active.
- (ii) A and B are enantiomers and product is optically inactive.
- (iii) Racemisation occurs and the product is optically inactive.

The correct statements are-

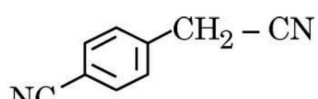
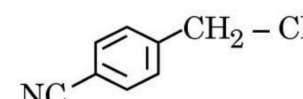
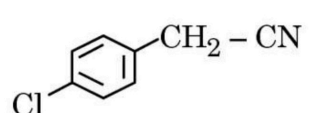
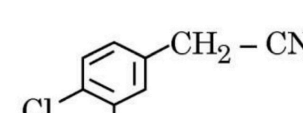
- (A) (i), (ii) and (iii)
- (B) (i) and (iii)
- (C) Only (iii)
- (D) (ii) and (iii)

11. Consider the following reaction:



1

The major product of the reaction is:

- (A)  (B) 
- (C)  (D) 

12. Alkyl halides undergoing bimolecular substitution reaction involve

1

- (A) Inversion of configuration
- (B) Formation of carbocation
- (C) Formation of racemic mixture
- (D) Retention of configuration



13. Which of the following is not an optically active compound? 1
- (A) Propan-2-ol (B) 2-bromopropanoic acid
(C) 2, 3-dihydroxypropanal (D) Bromochloro-iodomethane

For question numbers 12 and 13- two statements are given: one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions using the codes (A), (B), (C) and (D) as given below :

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.
12. Assertion (A) : The boiling points of alkyl halides decrease in the order : $RI > RBr > RCl > RF$.
Reason (R) : The boiling points of alkyl chlorides, bromides and iodides are considerably higher than that of the hydrocarbon of comparable molecular mass. 1
13. Assertion (A) : n-Butyl chloride has higher boiling point than n-Butyl bromide.
Reason (R) : C — Cl bond is more polar than C — Br bond. 1

SECTION - B

This section contains 4 Very Short Answer (VSA) type questions carrying 2 marks each.

14. A compound (A) with molecular formula C_4H_9I which is a primary alkyl halide, reacts with alcoholic KOH to give compound (B). Compound (B) reacts with HI to give (C) which is an isomer of (A). When (A) reacts with Ana metal in the presence of dry ether, it gives a compound (D), C_8H_{18} , which is different from the compound formed when n-butyl iodide reacts with Sodium. Write the structures of (A), (B), (C) and (D). Write the chemical reaction when compound (A) is reacted with alcoholic KOH. 2
15. Predict the order of reactivity of following compounds in S_N1 and S_N2 reactions: 2
- (a) The four isomeric bromobutanes
(b) $C_6H_5CH_2Br$, $C_6H_5CH(C_6H_5)Br$, $C_6H_5CH(CH_3)Br$, $C_6H_5C(CH_3)(C_6H_5)Br$
16. (a) Why is sulphuric acid not used during the reaction of alcohols with KI? 2
(b) Haloalkanes react with KCN to form alkyl cyanides as main product while AgCN forms isocyanides as the chief product. Explain.



17. (a) Why is thionyl chloride preferred in the preparation of haloalkanes? 2
(b) Write a chemical equation for a method that can help detect the position of a double bond in a haloalkane. What does its synthesis result in?

SECTION - C

This section contains 5 Short Answer (SA) type questions carrying 3 marks each.

18. Why are aryl halides extremely less reactive towards nucleophilic substitution reactions? Explain giving examples. 3
19. Although chlorine is an electron withdrawing group, yet it is ortho-, para-directing in electrophilic aromatic substitution reactions. Why? 3
20. Write main products formed when: 3
- (a) Methyl chloride is treated with NaI/Acetone.
- (b) 2, 4, 6-trinitrochlorobenzene is subjected to hydrolysis.
- (c) n-Butyl chloride is treated with alcoholic KOH.
21. (a) Differentiate between enantiomers and chiral molecules. Give an example of each. 3
(b) What are some properties of enantiomers?
(c) Explain the process of racemisation.
22. (a) Comment on the reactivity of Vinylic and phenylic halides in S_N1 and S_N2 reactions. 3
(b) Why is Grignard reagent stored under anhydrous conditions?

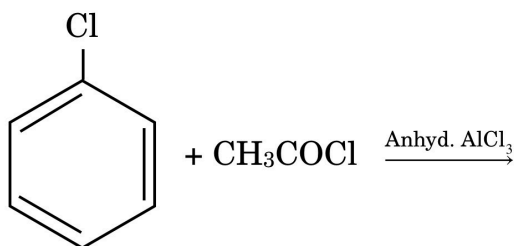
SECTION - D

This section consists of 1 case-based/source-based questions carrying 4 marks.

23. The polarity of C — X bond of alkyl halides is responsible for their nucleophilic substitution, elimination and their reaction with metal atoms to form organometallic compounds. Alkyl halides are prepared by the free radical halogenation of alkanes, addition of halogen acids to alkenes, replacement of O — H group of alcohols with halogens using phosphorous halides, thionyl chloride or halogen acids. Aryl halides are prepared by electrophilic substitution of arenes. Nucleophilic substitution reactions are categorised into S_N1 and S_N2 on the basis of their kinetic properties. Chirality has a profound role in understanding the S_N1 and S_N2 mechanism. 4



- (i) What happens when bromobenzene is treated with Mg in the presence of dry ether?
- (ii) Which compound in each of the following pairs will react faster in S_N1 reaction with OH^- ?
- (a) $CH_2 = CH - CH_2 - Cl$ or $CH_3 - CH_2 - CH_2 - Cl$
- (b) $(CH_3)_3C - Cl$ or CH_3Cl
- (iii) Write the structure of the major products formed in each of the following reactions:
- (a) $CH_3 - CH(Br) - CH_3 + KOH \xrightarrow[\text{heat}]{\text{Ethanol}}$
- (b)



- (iv) Write the equations for the preparation of 1-iodobutane from
- (a) 1-chlorobutane
- (b) But-1-ene

SECTION - E

This section consists of 2 Long Answer (LA) type questions carrying 5 marks each.

24. (a) Why does the NO_2 group show its effect only at ortho- and para- positions and not at meta- position in the reaction involving the replacement of a halogen by hydroxyl group? 5
- (b) Explain the mechanism behind the above reaction.

25. (a) Identify A, B, C, D, E and F in the following: 5
- $$\begin{array}{ccccccc}
 & & & & & \xrightarrow{\text{alcoholic KOH}} & A \xrightarrow{\text{HBr}} B \\
 E \xleftarrow{H_2O} D \xleftarrow[\text{dry ether}]{Mg} & CH_3 - CH - CH_2 - Br & & & & & \\
 & | & & & & & \downarrow \text{Na/dry ether} \\
 & CH_3 & & & & & C \\
 & \downarrow NaOC_2H_5 & & & & & \\
 & F & & & & &
 \end{array}$$

- (b) Write the structure of the major alkene formed by β -elimination of 2,2,3-trimethyl-3-bromopentane with sodium ethoxide in ethanol.



(c) Which one of the compounds in the following pairs is chiral?

